

INFOTRONICS AND INFORMATICS: An Emerging Multi-Disciplinary Engineering Platform For Industrial Development



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The automobile industry is radically transformed by the intervention of emerging technologies in the same field. This has not only influenced the way automobiles are built, but also impacted the society. A concrete study on the same led to a better understanding of the industry, and it became easy for industries to link automobiles to their perspective owners.

Introduction of information and communication technology (ICT) had a major role in reducing side effects like pollution and environmental hazards to the biosphere and making life safer by transforming the whole automobile sector from product to service industry.

Main focus of this study is on the enterprise transformation of the automobile industry due to the introduction of ICT, electronics and design concepts in India. Major areas of concern where we experienced a big leap are protection of wealth, regulations, safety, psychological comforts, smart communication, motivational, finance, marketing and so on.

In addition to this, a detailed educational and research opportunity at the national level is projected in the area of automotive informatics, infotonics, design, manufacturing, testing, certification and policy making. The present initiatives by educational institutes, the government of India, professional bodies as well as required futuristic interdisciplinary steps in courses to popularise automobile engineering are discussed here.

Initiatives by the government at the national level, universities and institutes at their level with a clear vision to uplift automobile domain with respect to facilities, research based functional areas and industry-institute linkage, collaborations and contributions to skilled human resources,

fast industry growth, customer security and national growth are also outlined.

A paradigm shift in domains

The objective of this study is to review and understand the role of electronics, communication and information technology (IT) in the automobile industry and its transformation from the last hundred years, when work was manually done, to the era of computers, when advanced computing techniques are used for data storage and energy to minimise time and labour to a notable extent.

When we compare non-digital IT with the existing smart management information system (MIS), it gives a clear understanding on how the industry has evolved in the last few decades. Here, we analyse various paradigm shifts in important domains in the field of automobile engineering with a focus on automotive infotonics and informatics.

Safety of property and regulations. Earlier, the automobile industry and its products were confined only to the upper class and was the most expensive possession after a house. Gradually, use of automobiles became common and even middle-class people started to own automobiles and as a result the increasing numbers became difficult to maintain on a registry of paper.

Without registration, an automobile could not be located after being stolen. Risks associated became a point of serious concern when immature drivers drove the vehicles and led to casualties and threats. Hence, it was an important necessity to put forth a flawless registration system along with legitimating the owners and allowing driving only if they hold a valid driving licence.

Moreover, as people got transferred from one place to another, vehicle registration